INCH-POUND

A-A-52114A
September 30, 1993
SUPERSEDING
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September 30, 1992

COMMERCIAL ITEM DESCRIPTION

MATTAX, WITH CARRIER

The General Services Administration has authorized the use of this Commercial Item Description in preference to type II, class D of Federal Specification GGG-H-506.

Abstract. This commercial item description covers a dual purpose tool combining the functions of an ax and a mattock and is intended for use in hand excavation. The wooden handle is removable for separate storage in the carrier.

1. SALIENT CHARACTERISTICS

1.1 Description. The mattax consists of three components as specified below.

1.1.1 Head. The head shall be configured and dimensioned as shown on figure 1. The head shall weigh 2 ± 1/4 pounds. Both the mattock blade and the ax blade shall be evenly sharpened with a sharp cutting edge. The bevel on the mattock head edge shall be on the bottom of the blade. The head shall be made of a forged process using carbon steel. The hardness of the blades extending 1-1/2 inch minimum distance back from the cutting edges shall be 37 to 50 on the Rockwell Hardness "C" Scale or 344 to 481 Brinell Hardness Scale (see 2.5). Prior to painting, the head shall be mechanically cleaned by shot blasting or tumbling. Cleaned heads shall be painted with a lusterless commercial grade enamel and the color shall match Color Chip 34094 of FED-STD-595. Each head shall have identification marking etched or indented (stamped) in a permanent and legible manner with the manufacturer's name or identification symbol, part number and country of origin.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Commander, US Army Natick Research, Development and Engineering Center, ATTN: SATNC-IRS, Natick, MA 01760-5019, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5120

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1.1.2 Handle. The handle shall be made from Grade A hickory in accordance with Federal Specification NN-H-93, so as to conform to the dimensions shown on figure 2. The top portion of the handle shall be tapered such as to provide a secure slip fit to the head. The bottom of the handle shall have a chamfered or beveled edge in accordance with Figure 2. Prior to painting, the handle shall be free from any contaminants (e.g., dust, dirt and oils). It shall be painted with a lusterless commercial grade enamel and the color shall match Color Chip 34094 of FED-STD-595. After painting, the handle shall have branded or indented (stamped) in a clear and legible manner. "CAUTION: PROTECTIVE EYEWEAR, GLOVES, AND HELMET MUST BE WORN DURING USE. DO NOT CUT HANDLE." The brand of indent shall be centered on the handle, aligned with the handle axis. The minimum lettering size shall be 1/4 inch.

1.1.3 Carrier. The carrier shall be constructed from cloth conforming to class 2 of MIL-C-43734, except that it shall be back coated and meet the requirements for hydrostatic resistance, stiffness, and spray rating. The carrier shall have a maximum weight of 10.5 ounces. The design shall be as shown on U.S. Army Natick drawings 2-2-506, 2-2-507, and 2-2-508. Webbing shall be sewn into the interior of the carrier at sections indicated on the government furnished patterns. The carrier shall have handle loops. The carrier shall have two keepers with slides for use as a means of attachment to a belt or strap. Two snap fasteners shall be located 1/8 inch from the top edge and 1/8 inch from side of the bottom flap, approximately 2-1/2 inches apart. The snaps shall be on the bottom (mattock side) to close the flap.

1.2 Thread. The thread used for seaming and stitching shall conform to type I, class 1, subclass B, size F, color Olive Drab S-1, C.A. 66022 of V-T-285. As an alternate, type I or II, class B, size F, color Olive Drab S-1, C.A. 66022 of V-T-295 may be used. All bartacks shall use the thread specified above, except that size E shall be used.

1.3 Webbing. The webbing used for the handle loops and for placement on the interior of the carrier shall conform to type I of MIL-W-43668. Webbing conforming to class 2, type VIIIa of MIL-W-4088 shall be sewn into the bottom for reinforcement for the mattock blade. The color of the webbing shall be Camouflage Green 483 of FED-STD-495 according to FED-STD-595.

1.4 Hardware.

1.4.1 Keeper with slide. The keeper with slide shall conform to type X, class B of MIL-H-9890.

1.4.2 Snap fastener. The snap fasteners shall conform to style 2, finish 2 of MIL-F-10884.

1.5 Seams and stitching. All seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, pleats, puckers loops, or open seams occur. Broken stitches shall be back stitched at least 1/2 inch. Thread

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tension shall be maintained so that no tight or loose stitching will result. Stitch type 301 (lockstitch) in accordance with FED-STD-751 shall be used for all seams on the carrier.

1.6 Patterns. The government shall furnish patterns for the carrier. The government patterns shall be used to make the contractor's working patterns. The government patterns shall not be altered in any way. The working patterns may be modified to accommodate automation; however, if the working patterns are modified, the design and measurements must be maintained as intended by government patterns.

1.7 Fit. All components, when assembled, shall fit properly. The handle shall fit into the head, with the bottom of the handle fitting into the top of the head, and no more than 1 inch of handle exposed through the top of the head. The head shall fit into the carrier, with the ax blade being inserted first. The handle shall fit into the carrier with no more than 4 inches of handle above the top handle loop.

1.8 Product demonstration model. When required, offerors shall submit two mattaxes of their intended production. The product demonstration model will consist of units independently developed. Product demonstration models will be analyzed and evaluated by the Government. Evaluation criteria and basis for award are contained in the solicitation. Product demonstration models will be retained by the Government. One product demonstration model approved by the Government under the resultant contract will serve as a manufacturing standard for the corresponding production items delivered under this contract. Offerors are advised that the product demonstration model does not relieve the successful offeror of their responsibility to perform in accordance with this Commercial Item Description.

2. QUALITY ASSURANCE

2.1 Inspection. Sampling shall be IAW MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes. The lot size shall consist of all items offered for acceptance at one time. The sample unit is one item.

2.1.1 Visual Inspection. The inspection level shall be level II with an Acceptable Quality Level (AQL) of 4.0 defects per hundred units.

2.1.2 Dimensional Inspection. The inspection level shall be level S-2, with an AQL of 4.0 defects per hundred units.

2.1.3 Hardness Testing. The inspection level shall; be level S-2, with an AQL of 2.5 defects per hundred units. Testing shall be IAW ASTM E18 or IAW ASTM E10.

2.1.4 Coating adhesion testing. The inspection level shall be level S-2, with an AQL 2.5 defects per hundred units. Testing shall be in accordance with 2.6.

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2.2 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

2.3 Metric product. Products manufactured using metric units will be considered on an equal basis with those manufactured using inch-pound units, provided that the product measurements fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376 and that all other requirements of this CID are met.

2.4 Defects. The mattax with carrier shall be examined for characteristics conforming to 1.1.1, 1.1.2, 1.1.3, and the following defects:

Head: Finish not smooth or not adherent; not clean; not free of burrs; any rough surfaces, slivers, cracks; not cleaned as specified; not color specified; dimensions not as specified; blade not evenly sharpened.

Handle: Not clean; not free from chips, scrapes, paint streaks or drips, paint not adherent; not color specified; dimensions not as specified.

Carrier: Any component part missing, distorted, or not as specified; any hole, cut, tear, mend or burn; any open seam; any bartack missing or misplaced; any thread not as specified; dimensions not as specified; not as specified in 1.1.3.

2.5 Hardness test. The hardness test shall be IAW ASTM E 18 or IAW ASTM E 10 using a 10mm Tungsten carbide ball. Two readings shall be made on the head. One reading, 1.5 inch in from the mattock blade edge, the other 1.5 inch in from the ax blade edge. Any reading not within specified hardness tolerances shall constitute a failure.

2.6 Coating adhesion test. The adhesion of lacquer, varnish or enamel paint shall be tested by applying and removing pressure sensitive tape over cuts made in the coating. The cuts shall be located on a flat area of the tool of sufficient size for the test. When multiple areas adequate for the test are available on the tool, the testing location shall be varied from one test sample to the next. The test procedure shall be as follows:

a. Three lines shall be cut completely through the coating. The three lines shall be 1/2 inch plus or minus 1/16 inch long and 1/16 to 3/32 inches apart.

b. Another four lines shall be cut completely through the coating overlapping and at 90 degrees plus or minus 10 degrees to the first set of lines. These four lines shall be 1/4 inch plus or minus 1/16 inch long and 1/16 to 3/32 inches apart.

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c. The cut pattern shall be six 1/16 to 3/32 inch approximate squares/rectangles.

d. A length of semitransparent tape having an adhesion strength of 40 plus or minus 2.5 ounce per inch of width shall be firmly pressed over the cut pattern. All air pockets under the tape shall be eliminated and the tape shall be firmly adherent to the coating.

e. After 90 seconds, plus or minus 30 seconds, the tape shall be stripped form the coated surface by rapidly pulling the tape back upon itself at 180 degrees plus or minus 30 degrees.

f. The removal of one of more squares/rectangles of coating shall be cause for rejection.

2.7 In-process crack detection test. After heat treating, all mattax heads shall be 100 percent in-process tested for cracks. The test shall consist of a bidirectional wet magnet particle inspection in accordance with ASTM E 709. If a mattax head has one or more surface cracks, the mattax head may be reworked by grinding or buffing to eliminate the surface crack(s); but must maintain the overall weight, shape, and dimensional requirements.. All reworked mattax heads shall be retested. All mattax heads presented for acceptance shall have no surface cracks.

2.8 Fit examination. The components shall be assembled as specified in 1.8 and examined for the following defects: carrier too small, i.e., head fails to fit properly into carrier; carrier too big, showing excessive fullness or wrinkles; more than 4 inches of handle above handle loop when inserted; handle not fitting properly into head; more than 1 inch of handle above top of head.

3. Regulatory requirements. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practicable.

4. PACKAGING

4.1 Overseas.

4.1.1 Packaging. Each head shall be secured into the carrier. The handle shall be attached through the two handle loops. The complete mattax shall be packaged in a fiberboard box with inside dimensions of approximately 24-1/4 by 4-1/2 by 3 inches. Replacement handles and carriers need no packaging.

1.1.2 Packing. Six mattax, thirty replacement handles, or sixty replacement carriers shall be packed in a fiberboard box conforming to style RSC-L, grade 275 of PPP-B-636. The inside dimensions of the mattax box shall be approximately 24-1/2 inches in length, 9-1/2 inches in width and 9-1/2 inches in depth. The inside dimensions of the box for the mattax handles shall be

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approximately 24-1/4 inches in length, 19 inches in width, and 6 inches in depth. The inside dimensions of the box for the carriers shall be approximately 12-3/4 inches in length, 11-3/4 inches in width, and 10 inches in depth. The box shall be closed in accordance with the appendix of PPP-B-636.

4.1.3 Palletization. Boxes shall be palletized on a 4 way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 35 for mattax, number 90 for handles and number 103 for carriers IAW the appendix of MIL-STD-147.

4.1.4 Marking. Marking shall be in accordance with MIL-STD-129.

4.2 Domestic.

4.2.1 Packaging, packing, palletization and marking. Packaging, packing, palletization and marking shall be in accordance with ASTM D 3951.

5. NOTES

5.1 Ordering data.

a. Title, number and date of this document.

5.2 Sources of documents.

5.2.1 Sources of government documents. Copies of military and federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

5.2.2 Sources of Non-Government documents.

ASTM D 3951 - Standard Practices for Commercial Packaging
ASTM E 10 - Hardness of Metallic Materials, Brinell
ASTM E 18 - Hardness of Metallic Materials, Rockwell Hardness and Rockwell Superficial
ASTM E 709 - Practice for Magnetic Particle Examination

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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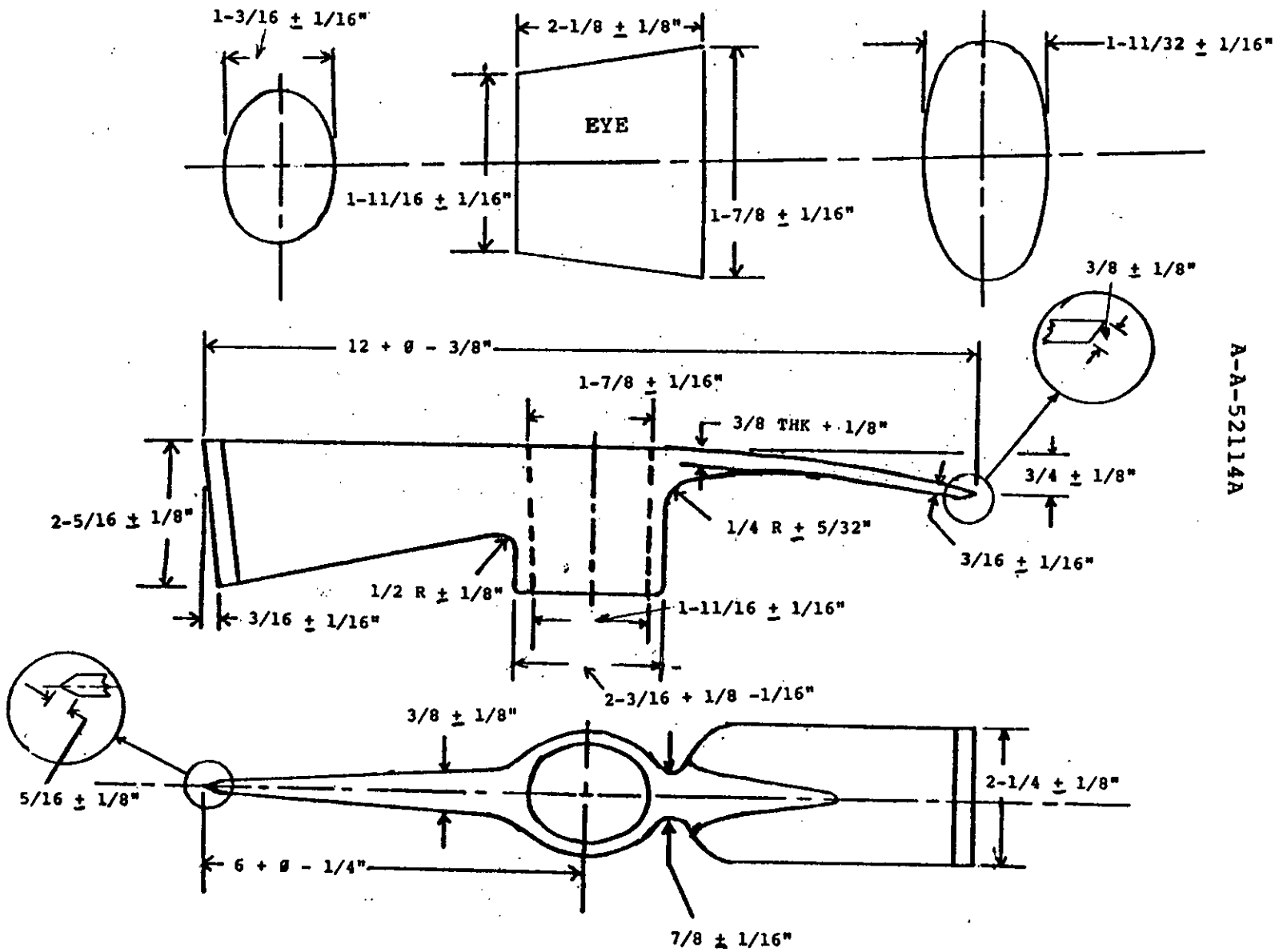


FIGURE 1. HEAD

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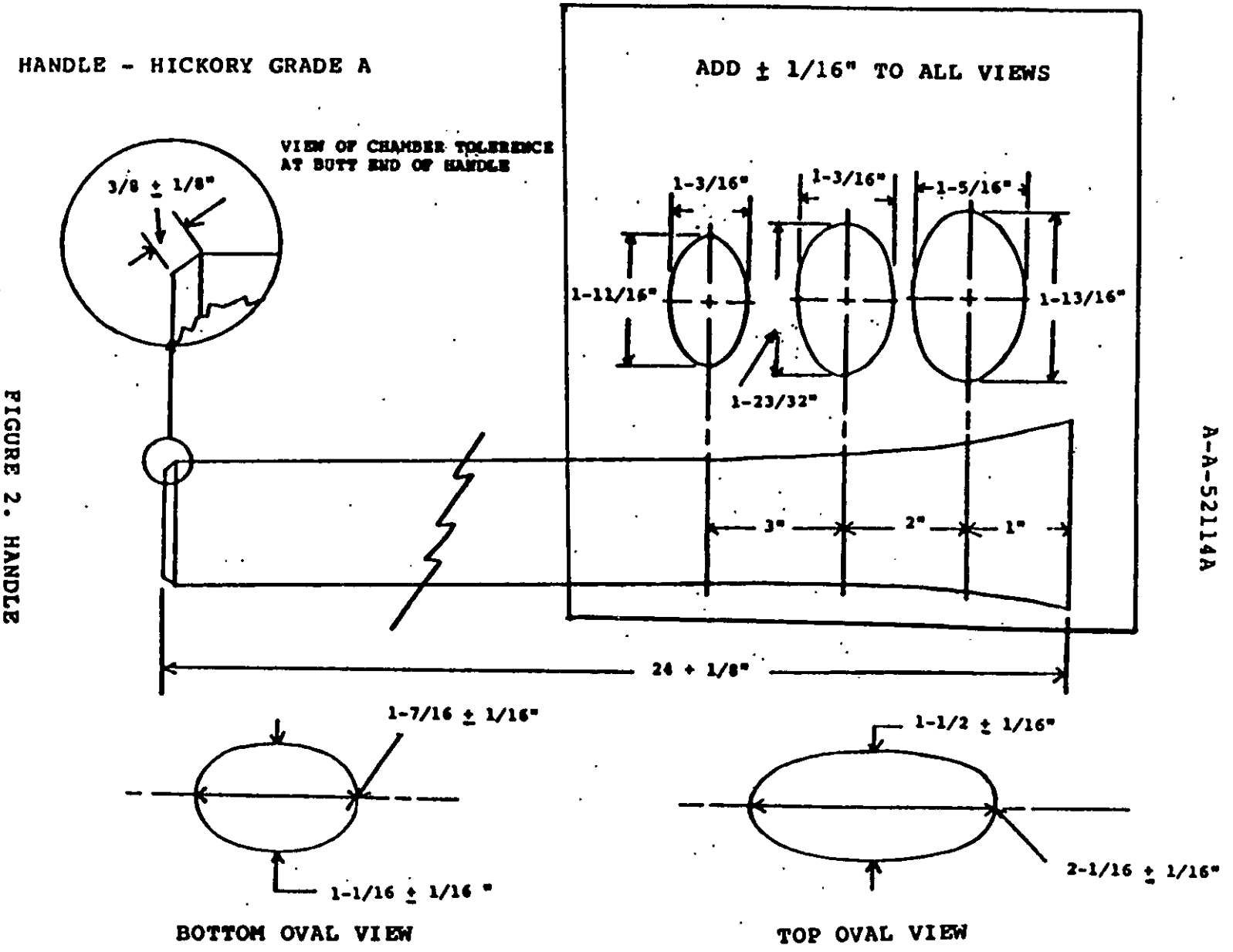


FIGURE 2. HANDLE

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MATTAX, WITH CARRIER

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME	b. TELEPHONE (Include Area Code)	
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